

### Instrument Items: Inclusion, Adaptation, and Justification

The following table presents all 30 items from the questionnaire (Section B: 10 items; Section C: 10 items; Section D: 10 items), along with the rationale for inclusion, any adaptations made from the original instruments, and the justification for each adaptation.

#### Section B: Attitudes Toward AI (Short GAAIS-10)

| Item Code      | Item Text                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------|
| B1<br>(Pos7)   | I am interested in using artificially intelligent systems in my daily physics learning.                  |
| B2<br>(Pos11)  | Artificial intelligence can have positive impacts on students' understanding of physics.                 |
| B3<br>(Pos12)  | Artificial intelligence is exciting for physics education.                                               |
| B4<br>(Pos17)  | Much of society will benefit from a future full of artificial intelligence in STEM fields.               |
| B5<br>(Pos18)  | I would like to use artificial intelligence in my own physics learning.                                  |
| B6<br>(Neg8)   | I find artificial intelligence sinister.                                                                 |
| B7<br>(Neg9)   | Artificial intelligence might take control of how students learn physics.                                |
| B8<br>(Neg10)  | I think artificial intelligence is dangerous for authentic physics learning.                             |
| B9<br>(Neg15)  | I shiver with discomfort when I think about future uses of artificial intelligence in physics education. |
| B10<br>(Neg19) | People like me will suffer if artificial intelligence is used more and more in physics learning.         |

**Section C: Trust in AI-EdTech (Adapted from Nazaretsky et al., 2025)**

| Item Code | Item Text                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 (PB1)  | AI can assist me in improving my responses to physics problems (e.g., open-ended questions, calculations) by suggesting personalised feedback in real time.                          |
| C2 (PB2)  | AI can assist me in self-learning physics by adjusting automatically individualised learning tasks adapted to my personal needs and level of knowledge.                              |
| C3 (PB3)  | AI (e.g., ChatGPT) can serve me as a learning pal or teaching assistant for physics.                                                                                                 |
| C4 (PB4)  | AI can help my physics teacher diagnose my strengths and difficulties so that teachers can assign individualised learning tasks adapted to my personal needs and level of knowledge. |
| C5 (PB5)  | AI can assist me with managing my physics learning activities (e.g., identifying if I am on track with my tasks, offering task prioritisation, optimising time management).          |
| C6 (PB6)  | AI can assist me in preparing for physics exams by identifying my knowledge gaps and providing relevant learning materials (e.g., explanations, practice problems).                  |
| C7 (PT1)  | As AI-based tools for personalised learning become more prevalent, I will trust them more for my physics learning.                                                                   |
| C8 (PT2)  | The more I know how an AI-based tool makes decisions, the more I can trust it for physics problem-solving.                                                                           |
| C9 (PT3)  | The more data is available for AI-based tools, the more I can trust its insights for physics concepts.                                                                               |
| C10 (PT4) | The more autonomy AI-based tools provide me to check or overwrite their automated decisions, the more I can trust them.                                                              |

**Section D: AI Anxiety (Shortened 10-Item Version)**

| Item Code    | Item Text                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| D1<br>(Q2)   | Learning to use AI techniques or products makes me anxious.                                                                          |
| D2<br>(Q4)   | Learning how an AI technique or product works makes me anxious.                                                                      |
| D3<br>(Q7)   | Being unable to keep up with the advances associated with AI techniques or products makes me anxious.                                |
| D4<br>(Q9)   | I am afraid that AI techniques or products will replace someone's job in the future.                                                 |
| D5<br>(Q15)  | I am afraid that widespread use of humanoid robots will take jobs away from people.                                                  |
| D6<br>(Q18)  | I am afraid that an AI technique or product may replace humans.                                                                      |
| D7<br>(Q11)  | I am afraid that if I begin to use AI techniques or products I will become dependent upon them and lose some of my reasoning skills. |
| D8<br>(Q44)  | I am afraid that an AI technique or product may get out of control and malfunction.                                                  |
| D9<br>(Q46)  | I am afraid of various problems potentially associated with an AI technique or product.                                              |
| D10<br>(Q45) | I am afraid that an AI technique or product may be misused.                                                                          |